

Model Predictive Control of Phase Shift Full-Bridge DC–DC Converter Using Laguerre Functions

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Abstract—The real-time computational load of an optimization problem plays a major role in the application of model predictive control (MPC) to fast switching power electronic converters. It is, therefore, highly desired to alleviate the computational burden of the MPC to render it feasible for these applications. In this brief, an efficient MPC algorithm based on Laguerre functions is proposed for a phase shift full-bridge (PSFB) dc–dc converter, in which the main control objective is to maintain the converter's load voltage at the desired set point while fulfilling multiple physical constraints, including the nonlinear peak input current constraint. It is shown in this work that the Laguerre functions' parameterization of MPC offers a promising solution to the high computational requirement associated with the conventional MPC without compromising on the dynamic closed-loop performance. The efficacy of the proposed control algorithm is tested on 60-W experimental hardware for 40- μ s sampling time. Experimental results are also compared with the conventional proportional-integral (PI) control structure in order to illustrate different aspects of the proposed control scheme.

Index Terms—DC–DC, full-bridge, Laguerre functions, model predictive control (MPC), nonlinear constraint, peak inductor current.

I. INTRODUCTION

PHASE-SHIFT full-bridge dc–dc converters are extensively employed in medium- and large-scale power applications for their distinct features—such as high-voltage conversion ratio, galvanic isolation, and flexibility of operation modes [1]–[3]. However, their successful deployment in these applications strongly predicates an exhaustive control design approach. The control design of the full-bridge converter, in particular, is an interesting and challenging task for the following reasons.

- 1) The system exhibits nonlinear open-loop dynamics, associated with different operating modes.
- 2) There are several linear and nonlinear physical constraints on states and control input of the system.
- 3) The physical system is frequently subjected to unmeasured input and output disturbances that need mitigation.
- 4) The computational time allowance for the control execution is of the order of microseconds.

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Many of the conventional linear control techniques, such as PID, state feedback, and LQR [4]–[6], can achieve a selection of the aforementioned objectives. Nevertheless, they are incapable to incorporate the ubiquitous operational constraints, such as the control input bounds and the peak current limitation [7], [8]. In such an application, model predictive control (MPC) is an excellent candidate due to its methodical approach to incorporate the physical constraints of the plant in the design process [9].

In the existing literature, the MPC of PSFB dc–dc converter has been discussed on a few occasions. A nonlinear constrained model predictive control for the PSFB dc–dc converter was initially proposed in [10] where the optimal control problem was solved with a sequential quadratic programming (SQP) algorithm. A nonlinear constraint of the peak input current was considered, and the experimental results for $N_p = 10$ were shown for the 10-kHz converter operation, where N_p is the prediction horizon. Later, another approach was proposed in [11] where linear MPC was shown to yield a similar computational performance as the nonlinear MPC [10] because of the hard nonlinear peak inductor current constraint. The experimental results were shown for the 300- μ s sampling interval. Recently, a heuristic nonlinear predictive controller approach has been proposed in [12] where it has been argued that the minimum-phase nature of the PSFB plant allows the use of a short prediction horizon, which, in turn, permits the converter operation on a 20-kHz sampling frequency for $N_p = 1$. A rather common observation about these approaches is that the high computational complexity associated with solving an optimization problem online results in a compromise on either the dynamic performance or the maximum switching frequency of the power-electronic converter. In most isolated dc–dc converters, up to 40% of the total converter volume is occupied by the magnetic components, such as the transformers [1], whose size varies inversely with the switching frequency. To this effect, it is desirable to lift the bound on the maximum switching frequency, so as to achieve the converter's power density objectives while maintaining a suitable dynamic performance in the presence of system constraints. This brief's contribution is given as follows:

- 1) to develop a low computation MPC design approach based on Laguerre functions to improve the switching frequency of the PSFB converter while making no compromise on the dynamic performance;
- 2) to make use of the system dynamics for transforming the nonlinear peak inductor current constraint into a dynamic linear constraint on the input control signal.

The system identification theory has benefitted from the use of orthonormal basis functions for many decades [13], [14]. In [15], it was argued that Laguerre functions can be used as an alternative parameterization for the discrete-time MPC problem as the optimal control sequence can be interpreted as the impulse response of a stable system. It was illustrated that, with only a small number of orthogonal basis functions, the optimal control input sequence for the discrete-time MPC control problem can be approximated with considerable precision. This feature of the Laguerre function-based approach can aid in significantly reducing the online computational burden of the optimal control problem in a practical application [16], [17]. Notwithstanding, to date, this scheme is underutilized and has not been harnessed to its full capability in real-world control applications. In this work, a Laguerre function-based MPC is designed for a PSFB dc–dc converter. It is argued that a dynamic performance similar to a conventional MPC with a long horizon can be attained with a considerably reduced computational burden, hence achieving a high-switching-frequency operation of the dc–dc converter. Moreover, the nonlinear peak current constraint has been quasi-linearized over the prediction horizon, considering that there is a significant time-scale separation between output voltage dynamics and the converter's sampling interval. The efficaciousness of the proposed control is experimentally validated on a 60-W lab-scale prototype of the PSFB dc converter with parameters highlighted in Table I. It is shown through several experimental results that the proposed control approach achieves an excellent closed-loop dynamic performance while successfully handling the stipulated physical constraints of the system. The experimental results are also compared with the conventional proportional–integral (PI) control structure to establish the feasibility of the proposed control algorithm.

The remainder of this brief is organized as follows. The circuit description and operation of the phase shift full-bridge (PSFB) dc–dc converter are discussed in Section II. Section III covers state-space modeling and constraints formulation of the plant. The Laguerre function-based MPC formulation is discussed in Section IV. The results of the proposed MPC control scheme and the PI control structure are discussed in Section V. The conclusions are drawn in Section VI.

II. DESCRIPTION AND OPERATION OF PSFB DC–DC CONVERTER

This section briefly describes the circuit configuration and working of the considered dc–dc converter for the discontinuous conduction mode (DCM).

A. Circuit Description

The configuration of the step-up PSFB dc–dc converter with its MPC control structure is shown in Fig. 1. The converter is powered by an input dc voltage source V_i with its current denoted by i_i . The input-side full-bridge circuit consists of two MOSFET legs indicated as A and B whose MOSFETs have been marked as S_1 – S_4 with their respective antiparallel diodes, D_1 – D_4 , respectively. The full-bridge circuit converts the input dc source voltage into ac and feeds it to the primary side of the

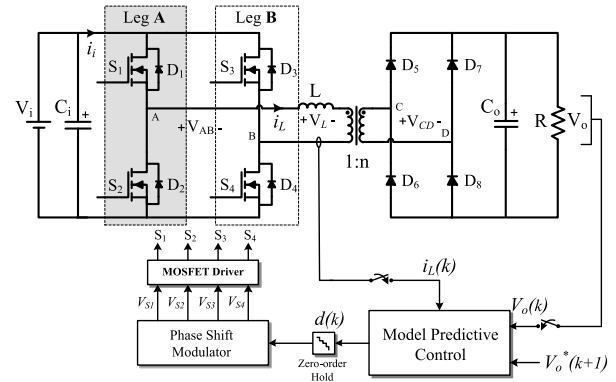


Fig. 1. Circuit illustration of the PSFC dc–dc converter with the model predictive control structure.

TABLE I
EXPERIMENTAL HARDWARE PARAMETERS OF PSFB DC–DC CONVERTER

Parameter	Notation	Value
DC Source Voltage	V_i	22V
Rated Output Voltage	V_o	60V
Turn Ratio	n	3.92
Output Capacitor	C_o	1990 μ F
Leakage Inductance	L	1.46 μ H
Rated Output Resistance	R	60 Ω
Switching Frequency	$f_s = 1/T$	25kHz
Rated Output Power	P	60W

step-up high-frequency transformer, with a turn ratio of 1 : n . The leakage inductance of the transformer is represented by L with its current denoted by i_L . The ac voltage on the secondary side of the transformer is rectified using the high-frequency diodes D_5 – D_8 and then filtered by an output capacitor C_o to produce a ripple-free output voltage V_o . Moreover, the output load resistance of the dc–dc converter is denoted by R .

The considered PSFB dc–dc converter works on the principle of phase shift pulsewidth modulation (PWM). In phase shift PWM, all the MOSFETs are driven with a constant switching frequency $f = 1/T$ and a fixed duty cycle of 50%. The MOSFETs in the same leg, such as S_1 and S_2 in leg A, operate with complementary PWM signals. This means that both switches cannot turn on at the same time. Similar is true for S_3 and S_4 in leg B. To control the voltage at the output, the phase shift θ between leg A and B PWM signals is varied such that $\theta \in [0, \pi]$. For the sake of generalization and convenience of the control design process, the phase displacement θ is normalized by π to yield $\theta = (dT/2)$, where T symbolizes the length of one switch interval, and d is the normalized control variable that ranges between 0 and 1 with $d = 0$, yielding the minimum output voltage, and $d = 1$, signifying the maximum voltage. From here on, d will be the manipulative variable in this control design problem.

B. DCM Operation of PSFB DC–DC Converter

For the existing control application, the designed PSFB dc–dc converter manifests DCM operation under the operating conditions of Table I. Therefore, only the DCM operation of the converter is discussed for brevity. The operational

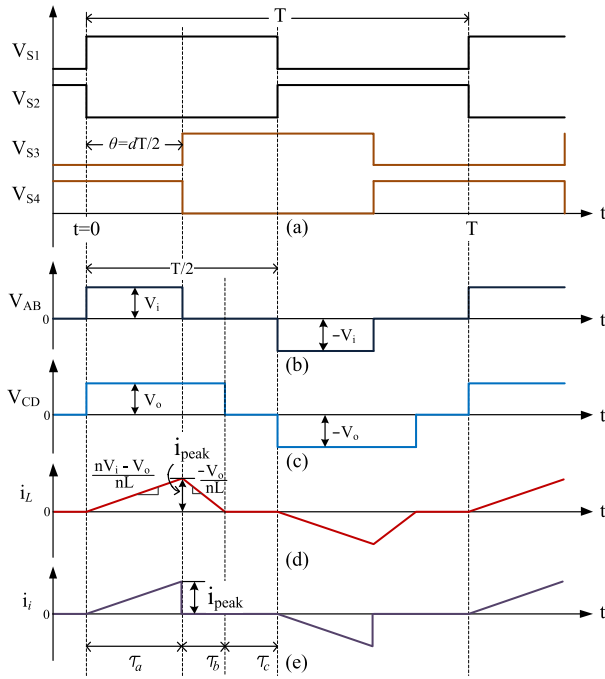


Fig. 2. PSFB dc-dc converter operational waveforms in DCM. (a) Phase shift PWM signals V_{S1} – V_{S4} . (b) Transformer primary-side voltage V_{AB} . (c) Transformer secondary-side voltage V_{CD} . (d) Inductor current i_L . (e) Supply current i_i .

waveforms of the PSFB converter in DCM are depicted in Fig. 2. The one half-cycle operation of the converter has been divided into three distinct intervals, namely, τ_a , τ_b , and τ_c . Fig. 2(a) shows the PWM signals V_{S1} – V_{S4} for switches S_1 – S_4 , respectively. The phase displacement between the legs A and B is $\theta = (dT/2) = \tau_a$. In the interval $t = 0$ – τ_a , the switches S_1 and S_4 conduct the forward current, and the transformer primary-side voltage V_{AB} is given by V_i , as shown in Fig. 2(b). The inductor current rises linearly and assumes its peak value i_{peak} at $t = \tau_a$, which is given as

$$i_{peak} = i_L(\tau_a) = \frac{(nV_i - V_o)dT}{2nL}. \quad (1)$$

In this interval, the power is supplied to the output port via D_5 and D_8 . Hence, the transformer secondary-side voltage V_{CD} becomes equal to V_o , as shown in Fig. 2(c). At $t = \tau_a$, V_{AB} drops to zero as a result of S_4 turning off. The inductor, however, keeps supplying power to the output as it is already charged. In the interval $t = \tau_a$ to $\tau_a + \tau_b$, the inductor current decreases linearly until it becomes zero at $t = \tau_a + \tau_b$, as shown in Fig. 2(d). At this instant, diodes D_5 and D_8 stop conducting, and V_{CD} also becomes zero.

In the interval $t = \tau_a + \tau_b$ to $T/2$, the inductor current stays at zero, and the output capacitor C_o , which is already charged under the effect of i_L , supplies the power to the output. After the completion of the positive half-cycle, i.e., $t = (T/2)$, a similar operation is carried out in the negative half-cycle with the inversion of polarities on currents and voltages. The corresponding waveforms of the negative half-cycle are shown in Fig. 2(a)–(e).

III. MODELING, CONSTRAINTS, AND STATE-SPACE AUGMENTATION

A. State-Space Modeling of Full-Bridge DC-DC Converter

The state-space averaged modeling of the PSFB dc converter, operating in DCM, is well documented in the literature [18]–[21]. In this work, a nonlinear averaged model of the full-bridge dc-dc converter proposed in [20] has been employed

$$\dot{\bar{i}}_L = \frac{dV_i}{L} - \frac{4\bar{i}_L\bar{V}_o}{dT(nV_i - \bar{V}_o)} \quad (2)$$

$$\dot{\bar{V}}_o = \frac{\bar{i}_L}{nC_o} - \frac{\bar{V}_o}{RC_o} \quad (3)$$

$$y' = \bar{V}_o \quad (4)$$

where $\bar{i}_L$ and $\bar{V}_o$ are the state variables representing the averaged inductor current and the output voltage, respectively, over a switching half-cycle, and y' is the measured output.

The open-loop plant (2)–(4) has been linearized across the nominal equilibrium point, $\bar{i}_L^* = 3.7$ A, $\bar{V}_o^* = 60$ V, and $d^* = 0.27$ for a nominal load resistance of 60Ω and is given by

$$\begin{aligned} \dot{x}_c &= A_c x_c + B_c u_c \\ y_c &= C_c x_c \end{aligned} \quad (5)$$

where

$$\begin{aligned} A_c &= \begin{bmatrix} \frac{-4\bar{V}_o^*}{d^*T(nV_i - \bar{V}_o^*)} & \frac{-4nV_i\bar{i}_L^*}{d^*T(nV_i - \bar{V}_o^*)^2} \\ \frac{1}{nC_o} & \frac{-1}{RC_o} \end{bmatrix}, \quad x_c = \begin{bmatrix} \bar{i}_L - \bar{i}_L^* \\ \bar{V}_o - \bar{V}_o^* \end{bmatrix}, \\ B_c &= \begin{bmatrix} \frac{V_i}{L} + \frac{4\bar{i}_L^*\bar{V}_o^*}{d^{*2}T(nV_i - \bar{V}_o^*)} \\ 0 \end{bmatrix}, \quad C_c = [0 \ 1], \quad u_c = d - d^*. \end{aligned} \quad (6)$$

Here, A_c , B_c , and C_c are the continuous-time system matrix and the input and output matrices of the linearized system, respectively. Correspondingly, x_c , u_c , and y_c denote the state vector, input vector, and output vector of the system.

B. Augmentation of State-Space Model for Offset-Free Tracking of Output

In dc-dc converters, the output load R is entirely user-defined and can change at any time. Therefore, a mechanism needs to be in place in order to reject any load disturbances in the system to ensure error-free reference tracking for the output voltage. This is achieved by constructing the plant model with incremental state variables rather than the actual state variables. First, the continuous state-space model of (5) and (6) is discretized using zero-order hold with sampling time T . The resulting discrete-time model is given as

$$\begin{aligned} x_m(k+1) &= A_m x_m(k) + B_m u_m(k). \\ y_m(k) &= C_m x_m(k). \end{aligned} \quad (7)$$

Here, A_m , B_m , and C_m are the discrete-time system matrices given by

$$A_m = e^{A_c T}, \quad B_m = \int_0^T e^{A_c \tau} B_c d\tau, \quad C_m = C_c \quad (8)$$

and $x_m(k)$ and $x_m(k+1)$ denote the state vectors corresponding to $t = kT$ and $t = (k+1)T$, respectively.

The integral action is embedded into the system by appending the output of the system to the original state vector. The new augmented model is given as

$$\begin{aligned} x(k+1) &= Ax(k) + B\Delta u(k), \\ y(k) &= Cx(k) \end{aligned} \quad (9)$$

where

$$A = \begin{bmatrix} A_m & O_{1 \times 2}^T \\ C_m A_m & I_{1 \times 1} \end{bmatrix}, \quad x(k) = \begin{bmatrix} \Delta x_m(k) \\ y_m(k) \end{bmatrix}, \quad (10)$$

$$B = \begin{bmatrix} B_m \\ C_m B_m \end{bmatrix}, \quad C = [O_{1 \times 2} \quad I_{1 \times 1}]. \quad (11)$$

In (9) and (10), $\Delta x_m(k) = x_m(k) - x_m(k-1)$, $\Delta u(k) = u_m(k) - u_m(k-1)$, and $y(k) = y_m(k)$, which is the displaced output voltage. Moreover, the matrices O and I represent zero and identity matrices, respectively.

C. Constraints

The PSFB dc–dc converter exhibits a number of operational and physical constraints that need to be fulfilled by the control system. The bounds on the manipulative variable d can be posed as linear constraints that are given by

$$0 \leq d \leq 1. \quad (12)$$

Furthermore, it is also desired to constrain the peak amplitude of the input current in order to protect the full-bridge semiconductor device setup from potential damage due to overcurrent. Since i_{peak} is not an averaged system state, the constraint is posed as a bound on the manipulative variable, such as

$$d_{\text{peak}} = \frac{2nLi_{\text{peak}}}{(nV_i - V_o)T} \quad (13)$$

where d_{peak} represents the value of the control input when the peak current constraint is active for a given V_o . Here, it is important to note that (13) is equally applicable in both the steady-state and transient modes of the converter operation. Fig. 3(a) shows the graphical representation of the constraint for $i_{\text{peak}} = 20, 30$, and 40 A, with d drawn as a function of V_o . Each curve signifies the limiting value of the control signal beyond which the constraint will be violated. It can be clearly seen in Fig. 3(a) that d_{peak} is a nonlinear function and, hence, cannot be posed as a linear constraint.

On the other hand, considering that the output V_o is measured at every sampling instant and that it has a very slow varying dynamics, i.e., $T \ll RC_o$, the constraint d_{peak} can be quasi-linearized at each sampling instant by placing the most recent measurement of V_o in (13) at each sampling interval. The resulting linearized constraint d_{peak}^* is given by

$$d_{\text{peak}}^* = d_{\text{peak}}|_{V_o=\tilde{V}_o} = \frac{2nLi_{\text{peak}}}{(nV_i - \tilde{V}_o)T} \quad (14)$$

where $\tilde{V}_o$ denotes the most recent measurement of V_o .

An important aspect of the existing application is that the model used for control design is applicable in the DCM only. Therefore, a boundary constraint has been devised to keep the dc–dc converter from jumping into CCM during

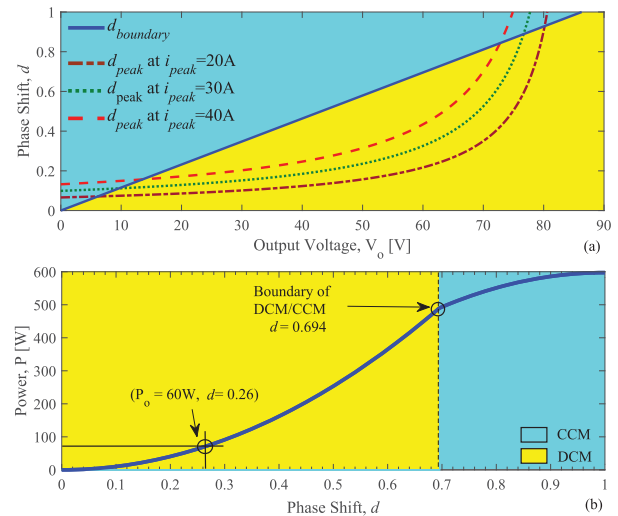


Fig. 3. (a) Constraint curves for peak inductor current d_{peak} for $i_{\text{peak}} = 20, 30$, and 40 A and DCM/CCM boundary (d_{boundary}). (b) Mean output power P drawn as a function of the control input d at $V_o = 60$ V.

nominal converter operation. Fig. 3(a) illustrates this constraint graphically as d_{boundary} , which is mathematically expressed as

$$d_{\text{boundary}} = \frac{V_o}{nV_i}. \quad (15)$$

Here, it is worthwhile to note that the steady-state operation of dc–dc converter at rated power is naturally carried out in DCM. Therefore, the deployment of (15) does not underutilize or impact the nominal converter operation even in transience. To endorse this, Fig. 3(b) depicts the mean output power P of the converter in terms of the phase shift d at $V_o = 60$ V. It can be seen that the converter switches from DCM to CCM at $P \approx 500$ W, which is significantly greater than the physical power rating of the dc converter, i.e., 60 W [10].

Finally, all the operational constraints of the PSFB dc–dc converters are written together as

$$\begin{cases} -d \leq 0 \\ d \leq 1 \\ d \leq d_{\text{peak}}^* \\ d \leq d_{\text{boundary}}. \end{cases} \quad (16)$$

IV. PROBLEM FORMULATION AND CONTROL DESIGN

Solving an MPC optimization problem in real time can be quite challenging in power electronic applications as the time constants are on a microsecond scale. It is, therefore, highly beneficial to alleviate the computational cost involved in solving the control problem.

A. MPC Control Formulation With Laguerre Functions

Historically, the Laguerre functions were mainly employed in the system identification theory to capture the impulse response of a system. As the control sequence of the MPC problem is essentially the impulse response of a stable system, it can be approximated using a finite quantity of orthogonal

basis functions [17]. A major advantage of this strategy is that, rather than solving for N_p variables as in conventional MPC, the coefficients of fewer basis functions are optimized to estimate the same control trajectory. This results in alleviating the computational cost of the MPC problem, which renders it highly feasible for practical applications, such as the PSFB dc-dc converter.

Consider the state-space model of the dc-dc converter (9) in the form

$$x(t_i + k + 1|t_i) = Ax(t_i + k|t_i) + B\Delta u(t_i + k|t_i). \quad (17)$$

At time t_i , the discrete-time control input trajectory is given by $\Delta U = [\Delta u(t_i) \Delta u(t_i+1) \dots \Delta u(t_i+N_p-1)]^T$. This control sequence can be posed as a weighted addition of a certain number of orthonormal basis functions given by

$$L(k) = [l_1(k) \ l_2(k) \ \dots \ l_N(k)]^T \quad (18)$$

where $L(k)$ denotes the matrix of these functions, and N is the enumeration of the orthogonal functions used in the parameterization. Moreover, each function $l_N(k)$ denotes the inverse z -transform of its respective Laguerre function $\Gamma_N(z, a)$, which is given by

$$\Gamma_N(z, a) = \left(\frac{z^{-1} - a}{1 - az^{-1}} \right)^{N-1} \frac{\sqrt{1-a^2}}{1 - az^{-1}} \quad (19)$$

where a represents the time factor or pole of the Laguerre network.

In the employed Laguerre parameterization, the differential control sequence $\Delta u(t_i + k|t_i)$ can be expressed as

$$\Delta u(t_i + k|t_i) = \sum_{\rho=1}^N c_\rho(t_i) l_\rho(k) = L(k)^T \eta \quad (20)$$

where c_ρ signifies the multiplier of its corresponding discrete Laguerre function $l_\rho(k)$ and η symbolizes the Laguerre coefficients vector that is given by

$$\eta = [c_1 \ c_2 \ \dots \ c_N]^T. \quad (21)$$

Substituting the value of $\Delta u(t_i + k|t_i)$ from (21) into (17), the parameterized state-space form is obtained in terms of η such that at time instant j

$$y(t_i + j|t_i) = CA^j x(t_i) + C\varphi(j)^T \eta \quad (22)$$

where

$$\varphi(j)^T = \sum_{\rho=0}^{j-1} A^{j-\rho-1} BL(\rho)^T. \quad (23)$$

Here, $L(k+1) = A_L L(k)$, and the base case is $L(1) = A_L L(0)$, where A_L signifies an $N \times N$ matrix that is a function of a and $\beta = 1 - a^2$ [15]. It is important to note here that the model is now formulated as a function of a new control input η instead of ΔU as in the traditional MPC approach.

In view of (17) and (21), the objective function J is defined over the prediction horizon N_p such that

$$J = \sum_{j=1}^{N_p} x(t_i + j|t_i)^T \bar{Q} x(t_i + j|t_i) + \eta^T \bar{R} \eta \quad (24)$$

where $\bar{Q}$ and $\bar{R}$ consist of weighting factors for the state and control variables, respectively. Making use of (22), the objective function of (24) is written as

$$J = \eta^T \Omega \eta + 2\eta^T \Psi x(t_i) + \sum_{j=1}^{N_p} x(t_i)^T (A^T)^j \bar{Q} A^j x(t_i) \quad (25)$$

where

$$\Omega = \sum_{j=1}^{N_p} \varphi(j) \bar{Q} \varphi(j)^T + \bar{R}; \quad \Psi = \sum_{j=1}^{N_p} \varphi(j) \bar{Q} A^j. \quad (26)$$

Assuming that Ω is full rank, the closed-form optimal solution η^* of (25) is given as

$$\eta^* = -\Omega^{-1} \Psi x(t_i). \quad (27)$$

After η^* has been obtained, the optimal control trajectory can be acquired as

$$\Delta U^* = [L(0)^T \eta^* L(1)^T \eta^* \dots L(N_p-1)^T \eta^*]^T. \quad (28)$$

Here, it can be effectively observed that the number of decision variables to compute the optimal control trajectory is N instead of N_p , which results in only a fraction of the computational burden compared to the traditional MPC approach.

B. Constrained Optimal Solution

Laguerre functions' parameterization of the MPC control problem also equips the designer with the flexibility to choose the future constraints' location in the prediction horizon.

For $m = 0, 1, 2, \dots, N_p - 1$, the constraints on the control input u_m are given by

$$u_{\min} \leq u_m(t_i + m) \leq u_{\max}. \quad (29)$$

$$u_{\min} \leq u_m(t_i - 1) + \sum_{m=0}^{N_p-1} \Delta u(t_i + m) \leq u_{\max} \quad (30)$$

where $u_{\min}$ and $u_{\max}$ represent the minimum and maximum control bounds, respectively, given by (16). Using (21), these constraints can be posed as a function of the new control input η as

$$\begin{cases} \tilde{M} \eta \leq U_{\max} - U_m(t_i - 1). \\ -\tilde{M} \eta \leq -U_{\min} + U_m(t_i - 1). \end{cases} \quad (31)$$

Here, $\tilde{M}$ is an $m \times N$ constraint matrix obtained using the interpolation of the Laguerre polynomial [14], while $U_{\min}$ and $U_{\max}$ are column vectors of dimensions $m \times 1$ consisting of $u_{\min}$ and $u_{\max}$ values evaluated in each sampling interval according to (16).

In order to obtain the constrained optimal solution, a matrix M_{act} is constructed with only the active inequality constraints. Then, a dual-quadratic program is solved using Hildreth's algorithm [22] to find the Lagrange multipliers λ_{act} [17], and the resulting constrained optimal solution is obtained by

$$\eta^* = -\Omega^{-1} (\Psi x(t_i) + M_{\text{act}}^T \lambda_{\text{act}}). \quad (32)$$

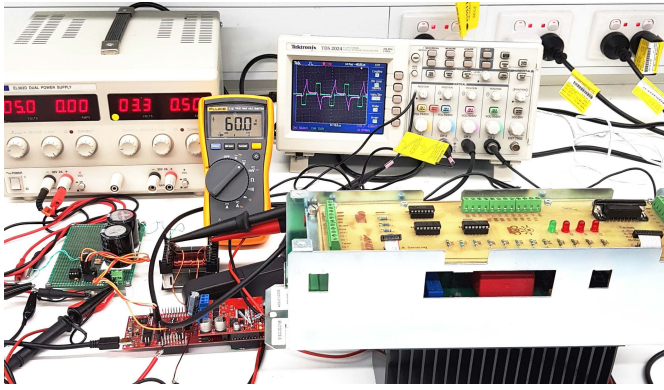


Fig. 4. Experimental hardware of the PSFB dc-dc converter setup with Texas Instrument F28069M DSP.

V. EXPERIMENTAL RESULTS

The proposed Laguerre function-based MPC algorithm has been experimentally validated on a 60-W lab-scale PSFB dc-dc converter hardware. The parameters of the experimental hardware are shown in Table I. In this control problem, only one of the $u_{\min}$ and $u_{\max}$ constraints can be active at a given time. Therefore, the parameter N is chosen as 3. Moreover, the pole of the Laguerre network is calculated as 0.8, based on the parameters of Table I [15]. In view of the computational performance, the constraint horizon has been chosen as three, which means that only the initial three control samples have been subjected to constraints, and the prediction horizon is chosen as 20. A Texas Instrument F28069M InstaSpin DSP hardware (90 MHz) has been used for the implementation of the MPC problem. It is estimated that the execution of the control problem takes nearly $33 \mu\text{s}$ in the worst case. The picture of the complete experimental hardware setup along with the DSP has been shown in Fig. 4.

A. Experimental Results With the Proposed MPC

1) *Startup*: The startup waveforms of the PSFB dc-dc converter with the proposed control algorithm have been shown in Fig. 5. For the startup process, the load resistance is taken as $R = 60 \Omega$, which signifies a full load of 60 W at rated operating conditions. The peak inductor current constraint is selected as 30 A. Fig. 5(a) shows that the output voltage rises from 0 to its nominal value of 60 V in nearly 150 ms. It can be observed in Fig. 5(b) and (c) that, for $V_o = 0$ to 10 V, the constraint of DCM/CCM boundary is active. This is because the PSFB converter tends to start with CCM as the output capacitor initially behaves like a short circuit and demands a large inrush current. In order to save the MOSFETs from permanent damage, the boundary constraint is enforced by the MPC controller, so as to provide a soft start at the beginning. Later, at $V_o = 10$ V, when the converter naturally operates in DCM, the peak current constraint is enforced to reach the desired output voltage as soon as possible. As seen in Fig. 5(b), the peak current remains constrained at 30 A. Fig. 5(c) shows the control input d during the startup process along with the constrained values of control input d_{peak}^* and d_{boundary} . It is evident that the proposed MPC controller successfully handles

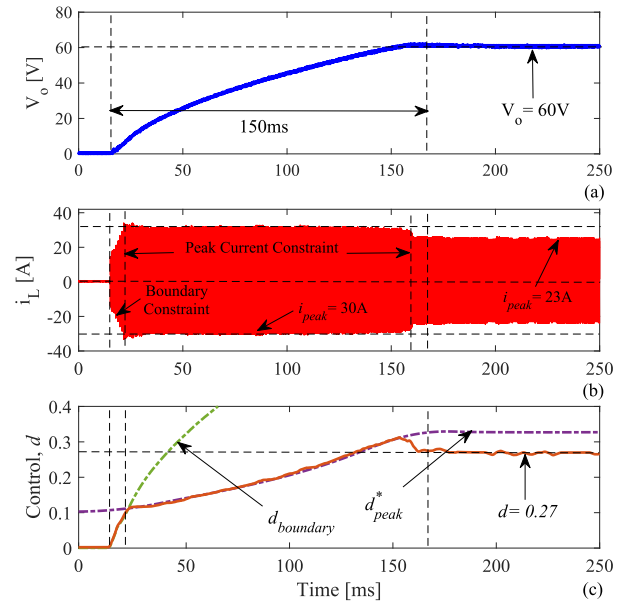


Fig. 5. Experimental waveforms of the PSFB dc-dc converter in startup with $R = 60 \Omega$ and $i_{\text{peak}} = 30$ A with the proposed MPC. (a) Output voltage V_o . (b) Inductor current i_L . (c) Phase shift d .

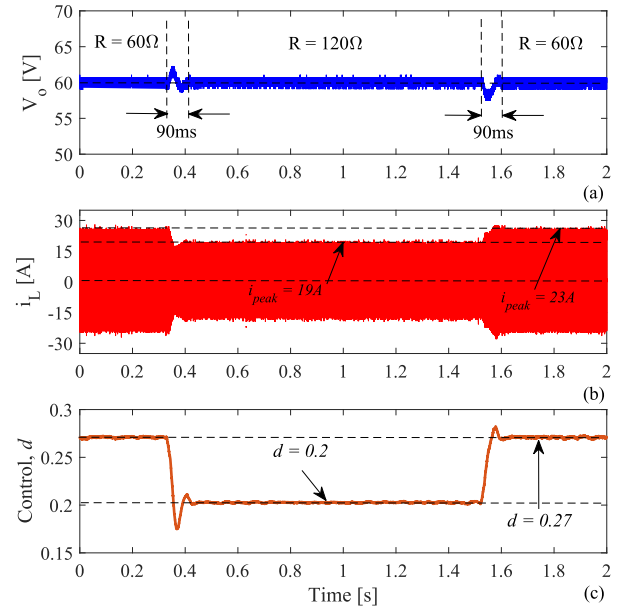


Fig. 6. Experimental waveforms of PSFB DC-DC converter for a step disturbance in load, i.e., $R = 60 \Omega \rightarrow 120 \Omega \rightarrow 60 \Omega$ with the MPC controller. (a) Output voltage V_o . (b) Inductor current i_L . (c) Phase shift d .

the peak current constraint with the mechanism proposed in Section III-C. Once the output voltage reaches 60 V, the peak current constraint becomes inactive, and the control input takes a steady-state value of 0.27.

2) *Load Disturbance Rejection*: The waveforms of the proposed MPC controller as a result of a load disturbance are shown in Fig. 6. At $t = 0.32$ s, R is doubled from 60 to 120 Ω . Fig. 6(a) shows that the controller successfully restores the desired steady-state voltage of $V_o = 60$ V within 90 ms and ≈ 2 V overshoot. Similarly, when the load resistance is reinstated as 60 Ω at $t = 1.52$ s, the controller rejects

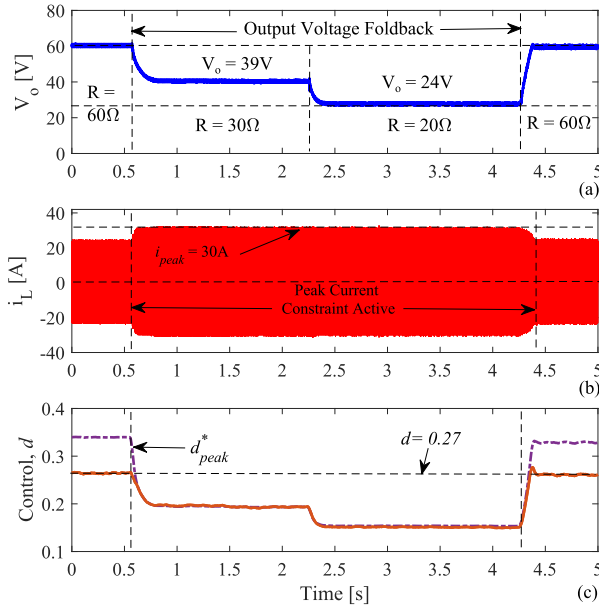


Fig. 7. Experimental waveforms for output voltage fold-back in the case of overload at the output, i.e., $R = 60\Omega \rightarrow 30\Omega \rightarrow 20\Omega \rightarrow 60\Omega$ with MPC controller. (a) Output voltage V_o . (b) Inductor current i_L . (c) Phase shift d .

the load disturbance again in nearly 90 ms with -1.5 -V undershoot. It is important to note here that the error-free reference tracking of the load voltage is achieved by means of the integrator embedded in the model structure.

3) *Output Fold-Back in Overload*: Fig. 7 shows the performance of the proposed MPC in case of overload at the output. At $t = 550$ ms, the output is subjected to an overload by halving the load resistance from 60 to 30Ω . In this case, the regulation of V_o at 60 V requires a $i_{peak} > 30$ A to maintain steady-state operation. However, since the maximum admissible peak current is 30 A, the constraint of peak current becomes active, as shown in Fig. 7(b). Consequently, the load voltage folds back from the rated 60 to 39 V, as shown in Fig. 7(a). At $t = 2.25$ s, the converter is pushed into worse overload by reducing R from 30 to 20Ω , which results in a further fold-back of V_o to 24 V. Then, at $t = 4.25$ s, the overload condition is removed by reinstating the load resistance as 60Ω . As soon as the overload is removed, the converter tends toward the normal operation and brings the output voltage back at 60 V. It is important to note that the fold-back mechanism works as self-defense against a potential overload on the converter output.

B. Experimental Results With PI Control

The results of Laguerre function-based MPC are compared with the widely applicable PI control scheme where the PI gains are calculated to yield the discrete-time poles of the system close to those with Laguerre MPC [23].

1) *Startup*: Fig. 8 shows the startup of the PSFB converter under full load with the PI controller. The maximum value of the control input for the antiwindup mechanism is selected as $d = 0.33$ instead of 1 in pursuit of avoiding the inrush current values that may possibly damage the hardware. As shown in Fig. 8(a), the output voltage rises to its desired reference

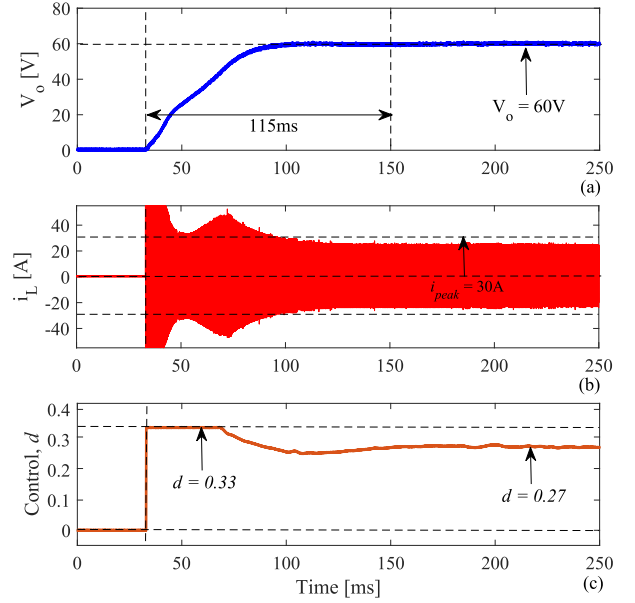


Fig. 8. Experimental waveforms of the PSFB dc-dc converter for startup with $R = 60\Omega$ using PI control. (a) Output voltage V_o . (b) Inductor current i_L . (c) Phase shift d .

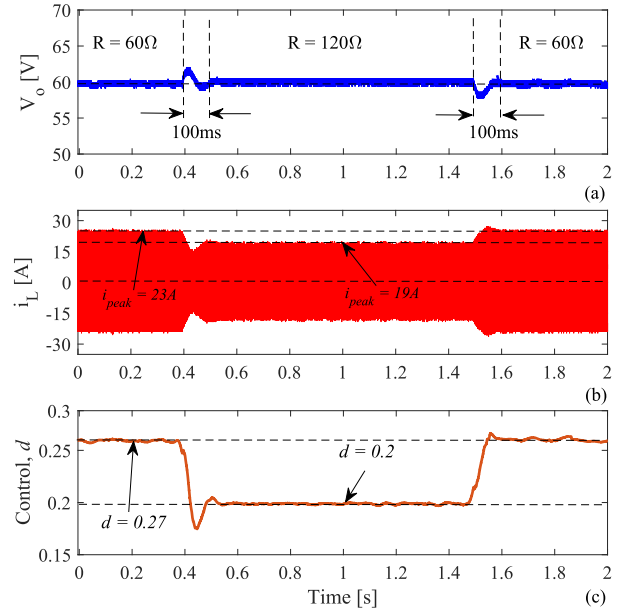


Fig. 9. Experimental waveforms as a result of step load disturbance, i.e., $R = 60\Omega \rightarrow 120\Omega \rightarrow 60\Omega$ with a PI controller. (a) Output voltage V_o . (b) Inductor current i_L . (c) Phase shift d .

of 60 V in nearly 115 ms. This is because an inrush current of > 55 A (≈ 83 A) is drawn at startup as the converter starts in CCM, as evident in Fig. 8(b). Then, at $V_o = 25$ V, the converter enters DCM as also predicted from Fig. 3(a), and the voltage rise becomes slower. It can be seen in Fig. 8(c) that the control signal remains saturated at 0.33 during the startup process, and eventually, the output voltage settles at 60 V and the control input at 0.27 . A very important thing to note here is that the faster rise time with PI control comes at the expense of potential damage to the hardware, which is not desirable in practical applications. Hence, a constrained

soft-start approach, such as MPC, is clearly a better choice from an application standpoint.

2) *Load Disturbance Rejection*: The load disturbance rejection waveforms of the PSFB converter for a step load change with the PI controller are depicted in Fig. 9. At $t = 0.4$ s, R is doubled from 60 to 120 Ω . As seen in Fig. 9(a), the PI controller rejects this disturbance in nearly 100 ms with an overshoot of ≈ 2 V in V_o , and the control input settles at 0.2. Then, at $t = 1.5$ s, R is halved from 120 to 60 Ω . Again, the load disturbance is entirely rejected by the PI controller in 100 ms with an undershoot of -2 V.

It can be noted from Figs. 6 and 9 that the proposed MPC exhibits a marginally better disturbance rejection compared to the PI control. In order to quantify the tracking performance, the mean squared error is calculated for the disturbance rejection of both the proposed MPC and PI. It is found that the proposed MPC produces an error of 3.6×10^3 compared to 4.03×10^3 with PI. Therefore, the proposed MPC control provides a superior overall control performance by successfully enforcing the stipulated operational constraints of the system.

VI. CONCLUSION

In this work, a computationally efficient MPC approach based on Laguerre functions has been proposed and applied to a PSFB dc–dc converter. A detailed discussion on the operation, modeling, and constraint formulation is included. The nonlinear constraint of the peak inductor current has been formulated as a dynamic linear constraint to make it suitable for the linear control formulation. The effectiveness of the proposed control scheme has been tested on a 60-W PSFB dc–dc converter hardware. It is illustrated through several results that the proposed Laguerre function-based MPC achieves an excellent dynamic closed-loop performance with three orthonormal basis functions while also successfully handling all constraints.

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